



ST1633i

Capacitive Touch Screen Controller

Datasheet

Version 1.5

2015/11/17

Note: Sitronix Technology Corp. reserves the right to change the contents in this document without prior notice. This is not a final specification. Some parameters are subject to change.

1 INTRODUCTION

The ST1633i is a capacitive sensing controller for small size projected capacitive touch screen. It is a RISC microcontroller with capacitor charge, capacitor sensing, slave I2C interface, general purpose I/O and embedded non-volatile memory.

Internal program and cooperating digital circuit convert finger / capacitor stylus physical touching into button pressing message or multiple coordination information for application. The maximum fingers identification ability is up to five.

The ST1633i uses low profile QFN package and support ITO electrode on glass or film substrate. Hence, slim and small touch panel module is realizable.

And more, low electromagnetic interference of ST1633i makes it suitable for modern touch screen application such as industrial equipment and automotive vehicle.

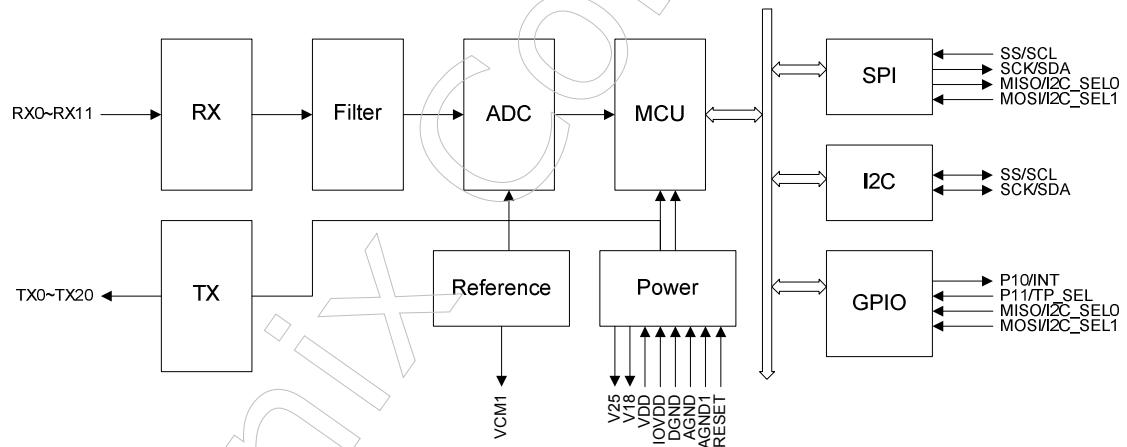
FEATURES

- MCU based touch controller
- Operation voltage
 - VDD = 2.7V ~ 3.6V
 - IOVDD = 1.6V ~ 3.6V
- Operation Temperature: -40°C ~ 85°C
- Storage Temperature: -40°C ~ 125°C
- Interface
 - I2C (slave)
 - SPI
- Sensor
 - 21TX, 12 RX
- Single finger handwriting
- five fingers detection and tracking
- Capacitive Sensor
 - Mutual-capacitance sensing
 - Report rate : 100Hz
 - Hardware noise reduction
 - Waterproof circuit

- Package
- QFN48

APPLICATIONS

- Appliance
- Medical
- Industrial automation
- Home automation
- Point of Sales
- Automotive



2 PACKAGE INFORMATION

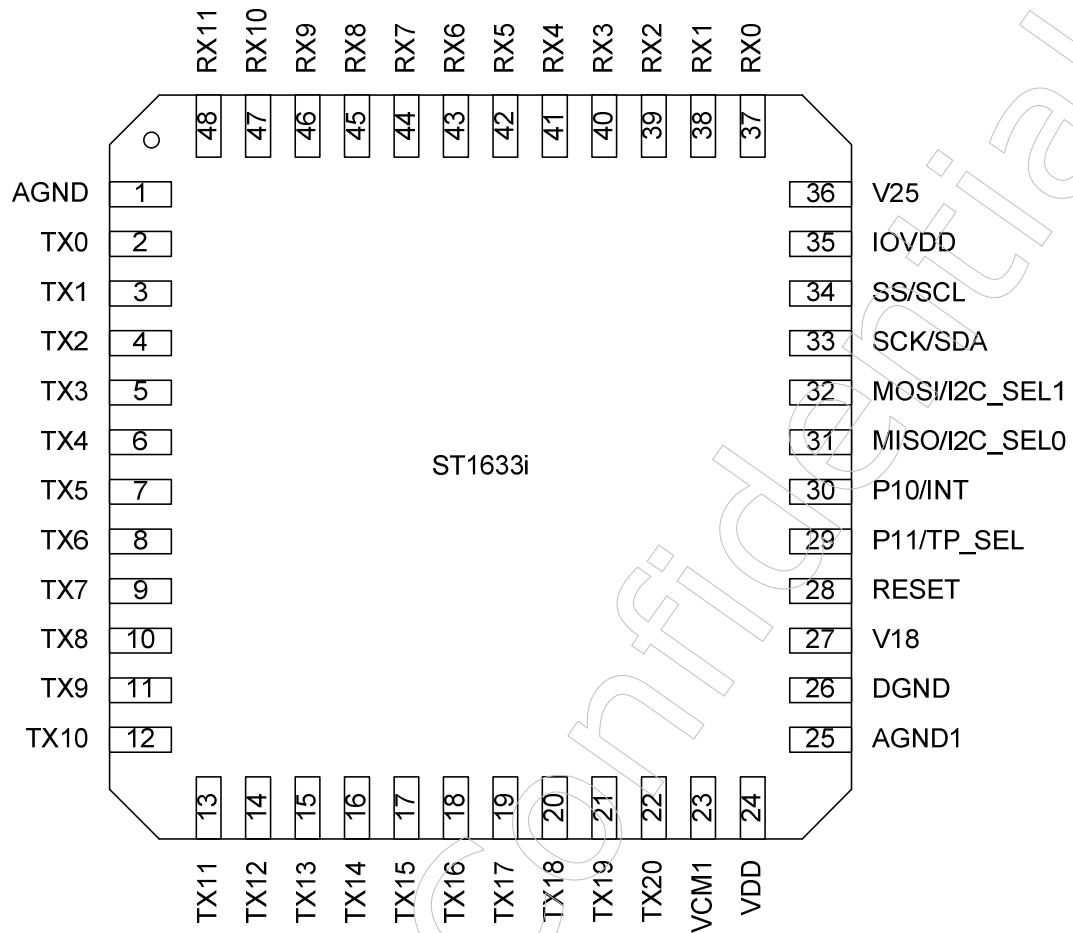


Figure 2-1 ST1633i Package Pin Configuration

Table 2-1 ST1633i Package Signal Descriptions

Pin #	Pin Name	I/O	Description
1	AGND	P	Analog ground
22~2	TX20~TX0	O	Transmitter channel
23	VCM1	P	Common mode voltage, connect to 1uF capacitor
24	VDD	P	Power supply, connect to 1uF capacitor
25	AGND1	P	Analog ground
26	DGND	P	Digital ground
27	V18	P	Digital power, connect to 1uF capacitor
28	RESET	I	System reset signal input, active low
29	P11/TP_SEL	I/O	P11 : General purpose I/O TP_SEL : Touch panel sensor select
30	P10/INT	I/O	P10 : General purpose I/O INT : Indicate coordinate data ready
31	MISO/ I2C_SEL0	I/O	MISO : SPI master input/slave output I2C_SEL0 : I2C address select
32	MOSI/ I2C_SEL1	I/O	MOSI : SPI master output/slave input I2C_SEL1 : I2C address select
33	SCK/SDA	I/O	SCK: SPI serial clock SDA: I2C serial data
34	SS/SCL	I/O	SS : SPI slave select SCL : I2C serial clock
35	IOVDD	P	I/O power supply, connect to 1uF capacitor
36	V25	P	Analog power, connect to 1uF capacitor
48~37	RX11~RX0	I	Receiver channel

Note: I/O type: P=Power pin, I=Input pin, O=Output pin

3 SYSTEM MANAGEMENT

3.1 Power Down

In power down mode, all of the clocks of ST1633i are stopped. The way to exit power down mode is by a hardware reset or I2C.

3.2 Reset

Master can reset ST1633i through RESET pin. RESET pin is low active and needs hold low for 1us to take effect.

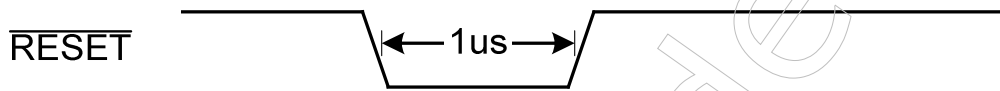


Figure 3-1 $\overline{\text{RESET}}$ Pin Low Pulse Width

3.3 Power On/Off Sequence

RESET pin should be held low before power on and power off. During power on, after both VDD and IOVDD reach normal voltage, RESET pin needs to be held low for 5ms to ensure internal block stable.

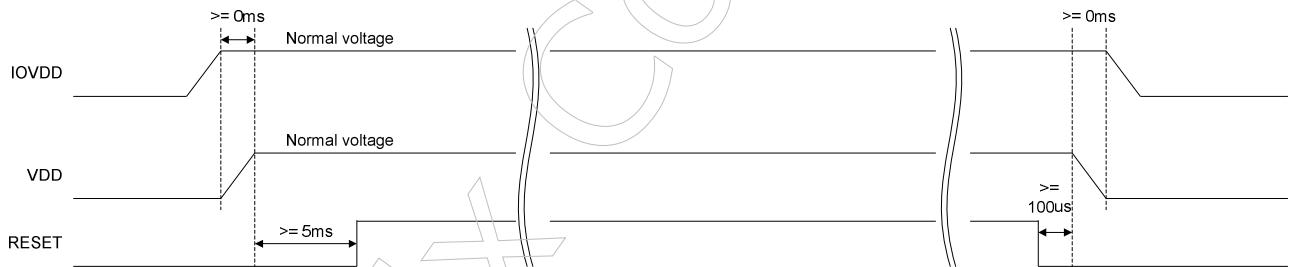


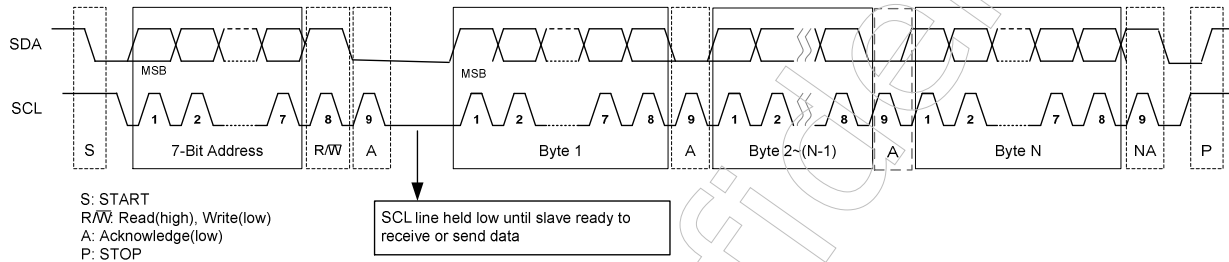
Figure 3-2 Power On/Off Sequence

4 DIGITAL INTERFACE

4.1 I2C Slave Interface

ST1633i equipped with I2C provides two wires, serial data (SDA) and serial clock (SCL), to carry transferring information at up to 400 kbit/s (Fast mode). ST1633i plays the slave role in I2C transfer. Both SDA and SCL are bidirectional lines, connected to IOVDD via pull-up resistors. All transactions begin with a START (S) and can be terminated by a STOP (P). 7-bits address follows START to recognize device. Each byte is 8-bits length and followed by an acknowledge bit. A HIGH to LOW transition on the SDA line while SCL is HIGH defines a START condition. A LOW to HIGH transition on the SDA line while SCL is HIGH defines a STOP condition. The data on the SDA line must be stable during the HIGH period of the clock. The HIGH or LOW state of the data line can only change when the clock signal on the SCL line is LOW.

Read



Write

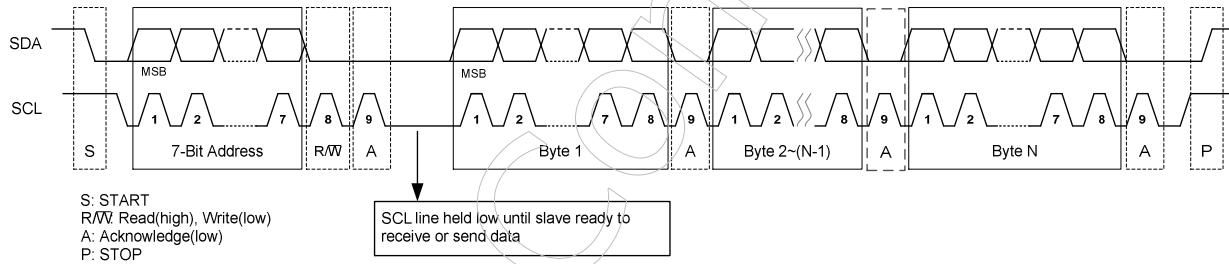


Figure 4-1 I2C Waveform

5 ELECTRICAL CHARACTERISTIC

5.1 Absolute Maximum Ratings

Table 5-1 Absolute Maximum Ratings

Parameter	Symbol	Min.	Max.	Unit
VDD	V_{VDD}	-0.3	+6	V
IOVDD	V_{IOVDD}	-0.3	+6	V
Operating Ambient Temperature	T_A	-40	+85	°C
Storage Temperature	T_S	-40	+125	°C

***Note:** Stresses above those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. All the ranges are stress ratings only. Functional operation of this device at these or any other conditions above those indicated in the operational sections of this specification is not implied or intended. Exposed to the absolute maximum rating conditions for extended periods may affect device reliability.

5.2 DC Electrical Characteristics

Table 5-2 System DC Electrical Characteristics

Condition: VDD = IOVDD = 3.3V, $T_A = 25^{\circ}\text{C}$, unless be specified individually.

Parameter	Symbol	Min.	Typ.	Max.	Unit	Condition
VDD	V_{VDD}	2.7	-	3.6	V	
IOVDD	V_{IOVDD}	1.6	-	3.6	V	
Operating Current	I_{NML}	-	16.1	24	mA	21TX, 12RX
Idle Current	I_{IDLE}	-	8.1	12.2	mA	21TX, 12RX, scan rate=20Hz
Power Down Current	I_{PD}	-	-	20	uA	
Input High Voltage	V_{IH}	0.85* I_{OVDD}	-	-	V	
Input Low Voltage	V_{IL}	-	-	0.15* I_{OVDD}	V	
Input Pull Up Resistor	R_{PU}	50	-	60	KOhm	
Output Driving Current	I_{DRV}	6	-	-	mA	$V_{OH} = IOVDD \times 0.8$
Output Sinking Current	I_{SINK}	10	-	-	mA	$V_{OL} = IOVDD \times 0.2$
Low Voltage Reset	V_{LVR}	-	-	2.3	V	

5.3 AC Electrical Characteristics

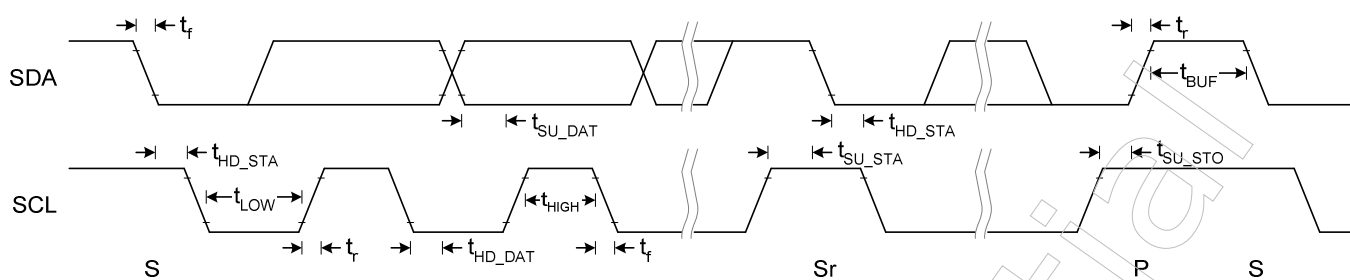


Figure 5-1 I2C Fast Mode Timing

Table 5-3 I2C Fast Mode Timing Characteristic

Conditions: VDD = 3.3V, GND = 0V, T_A = 25°C

Symbol	Parameter	Rating			Unit
		Min.	Typ.	Max.	
f _{SCL}	SCL clock frequency	0	-	400	kHz
t _{LOW}	Low period of the SCL clock	1.3	-	-	us
t _{HIGH}	High period of the SCL clock	0.6	-	-	us
t _f	Signal falling time	-	-	300	ns
t _r	Signal rising time	-	-	300	ns
t _{SU_STA}	Set up time for a repeated START condition	0.6	-	-	us
t _{HD_STA}	Hold time (repeated) START condition. After this period, the first clock pulse is generated	0.6	-	-	us
t _{SU_DAT}	Data set up time	100	-	-	ns
t _{HD_DAT}	Data hold time	0	-	0.9	us
t _{SU_STO}	Set up time for STOP condition	0.6	-	-	us
t _{BUF}	Bus free time between a STOP and START condition	1.3	-	-	us
C _b	Capacitive load for each bus line	-	-	400	pF

6 APPLICATION CIRCUITS

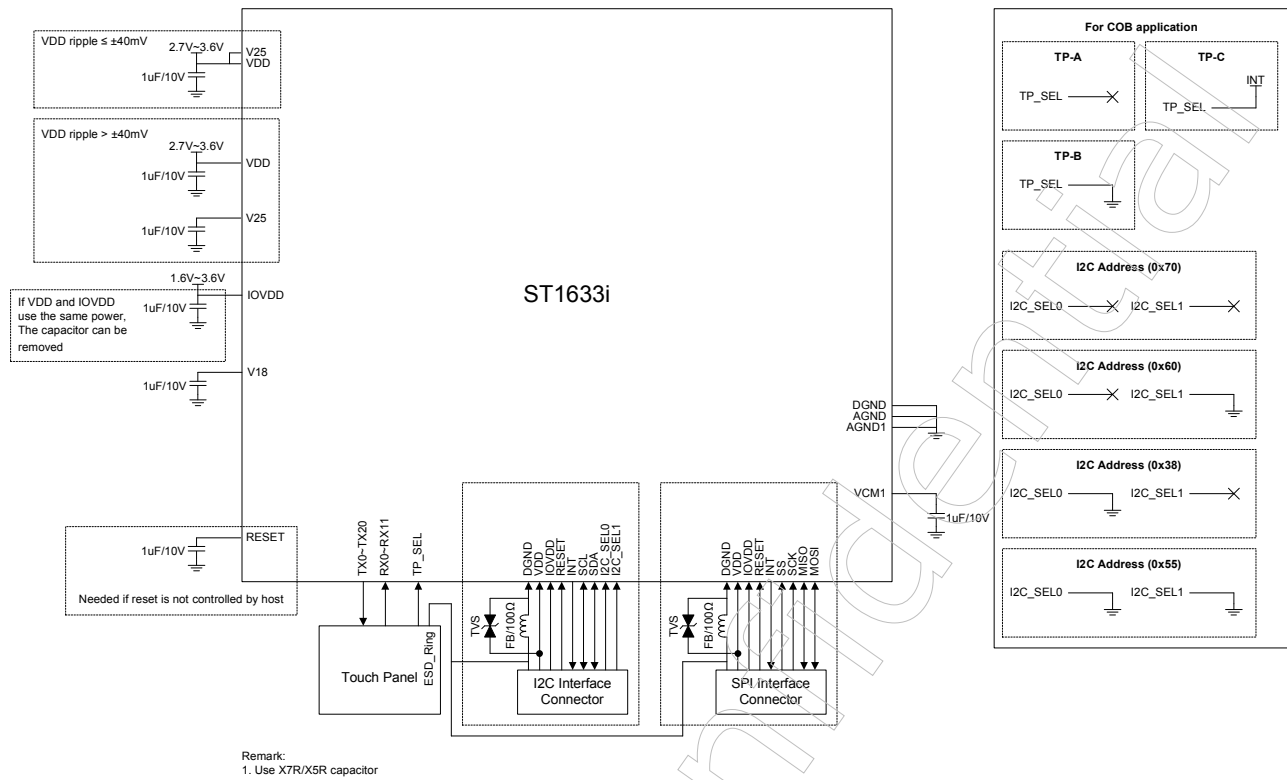
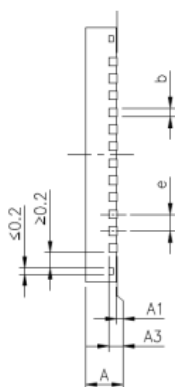
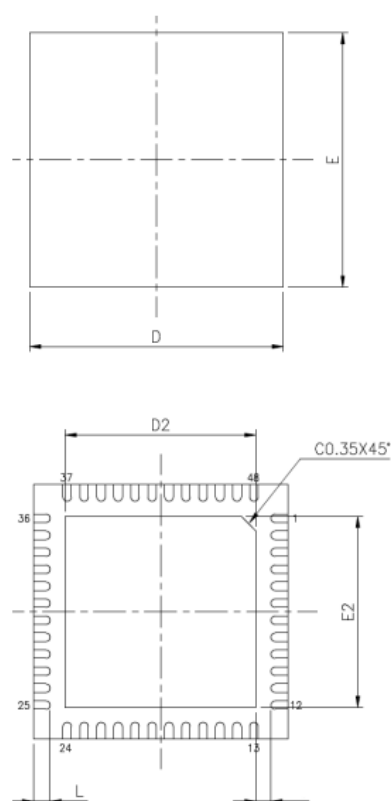


Figure 6-1 ST1633i Application Circuit

7 PACKAGE DIMENSION



SYMBOL	MIN.	MAX.
A	0.50	0.60
A1	0.00	0.05
A3	0.152 REF.	
b	0.15	0.25
D	6.00 (BSC)	
D2	4.57	4.72
E	6.00 (BSC)	
E2	4.57	4.72
L	0.35	0.45
e	0.40 (BSC)	

UNIT : mm

Figure 7-1 ST1633i Package Dimension

8 REVISION

REVISION	DESCRIPTION	PAGE	DATE
1.5	■ Remove "max. loading: 30 kOhm/60pF" in feature	3	2015/11/17
	■ Remove "IOVDD=3.3V" condition in VIH/VIL DC characteristic	8	
	■ Add two different V25 circuit in application circuit	10	
	■ Add "For COB application" note in application circuit	10	
1.4	■ Modify application circuit , revise TVS circuit	10	2015/01/13
1.3	■ Modify introduction section description	2	2014/12/31
	■ Modify applications feature	3	
	■ Add block diagram	3	
	■ P11 renamed to P11/TP_SEL, MISO renamed to MISO/I2C_SEL0, MOSI renamed to MOSI/I2C_SEL1	4,5	
	■ Add power on/off sequence	6	
	■ Modify I/O driving current to 6mA, sinking current to 10mA	8	
	■ Modify application circuit	10	
1.2	■ Delete schematic example section	-	2014/10/22
	■ Modify operating temperature from -20~80 degree to -40~85 degree	3,8	
1.1	■ Add operating/idle current data	8	2014/07/14
1.1	■ Modify operating temperature from -30~70 degree to -20~80 degree	3,8	2014/07/14
1.0	■ First release	-	2014/05/27

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